

**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND
TECHNOLOGY, KUMASI
COLLEGE OF ENGINEERING**

**BSc. (Electrical, Telecommunication, Computer and Biomedical Engineering)
MID Sem Examination 2020)**

**EE 156 : ELECTRICAL ENGINEERING DRAWING
(First Year)**

MARCH, 2020

Time allowed: ONE HOUR

INSTRUCTIONS FOR CANDIDATES:

1. Answer **ALL** questions
2. Circle the letter corresponding to your answer to a given question on the paper and shade the same letter on the scannable sheet provided
3. *Indicate your **Centre Name / Room number** on top of the Answer booklet*

INDEX NUMBER: 8265919

1. An on-delay timer has a pre-set time of 10 seconds. What happens when its coil is energised?
Its normally open contacts will
a) immediately close b) remain open **c) will close after 10second** d) remain close
2. An off-delay timer has a pre-set time of 10 seconds. What happens when its coil is energised?. Its normally open contacts will
a) immediately close b) remain open c) will close after 10second d) remain close
3. An off-delay timer has a pre-set time of 10 seconds. What happens when its coil is de-energised?
Its normally open contacts will
a) immediately close b) remain open **c) will close after 10second** d) remain close
4. Delay On De-Energized refers to which type of a timer?
a) OFF-Delay Timer b) Pulse Timer c) ON-Delay Timer d) Retentive ON-Delay Timer
5. The symbol shown in figure 1 is
a) an overload coil b) an overload contact c) a fuse link d) a disconnect switch

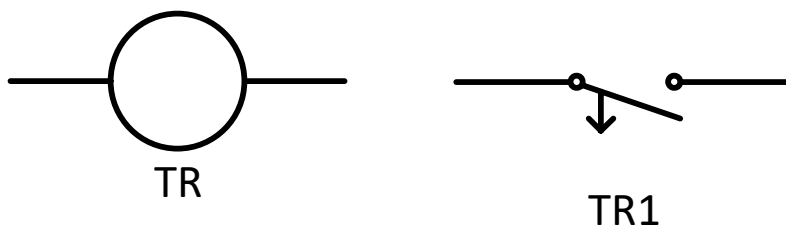
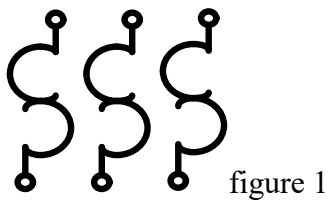


Figure 2

The figure 2 above shows a coil TR and its corresponding contact TR1 .

Use the information to answer questions 6 to 8

6. The system shown in figure 2 above is a/an
a) On-delay timer **b) Off-delay time** c) Overload relay d) Control relay
7. In figure 2, if the coil T1 is energized what happens to T2?

a) Nothing happens

c) T2 closes after a delay

b) T2 immediately closes

d) T2 becomes Normally Closed

8. In figure 2, if the coil T1 is de-energized

a) Nothing happens

c) T2 opens after a delay

b) T2 opens immediately

d) T2 becomes Normally Closed

9. In a motor control circuit, the stop button is usually

a) a normally closed push button

c) a selector switch

b) a normally opened push button

d) a toggle switch

Figure 3 below shows symbols (A to F) that are used to represent certain switches in motor control schematic diagrams. *Use it to answer questions 56 - 58*

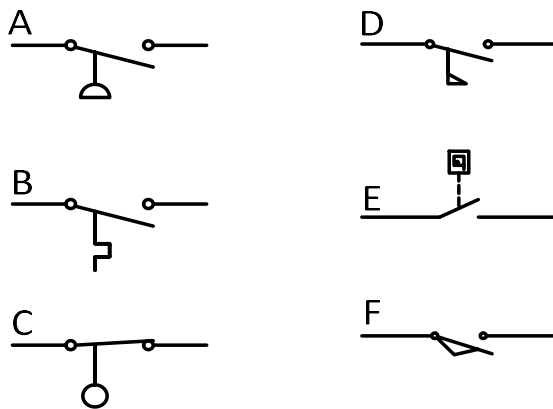


Figure 3

10. Which symbol in figure 3 represents a switch designed to pass electrical signal only when a predetermined limit is reached?

a) D

b) F

c) B

d) C

11. The temperature actuated switch in figure 10 is

a) A

b) E

c) D

d) C

12. Which switch in figure 10 monitors the direction of movement of liquid or gas substances

a) D

b) F

c) A

d) E

Figure 4 below is a control circuit diagram for a pump that is filling a tank with water.

Use it to answer questions 13-20.

13. Which of the following devices comes on when the tank is empty?

- a) Red lamp b) Amber lamp c) Motor **d) Both a) and c)**

14. When the red lamp is on, it indicates that

- I) The tank is full II) The tank is not full III) The motor is on IV) The switch FS is closed
a) I and III only **b) II and III only** c) III and IV only d) I, III and IV only

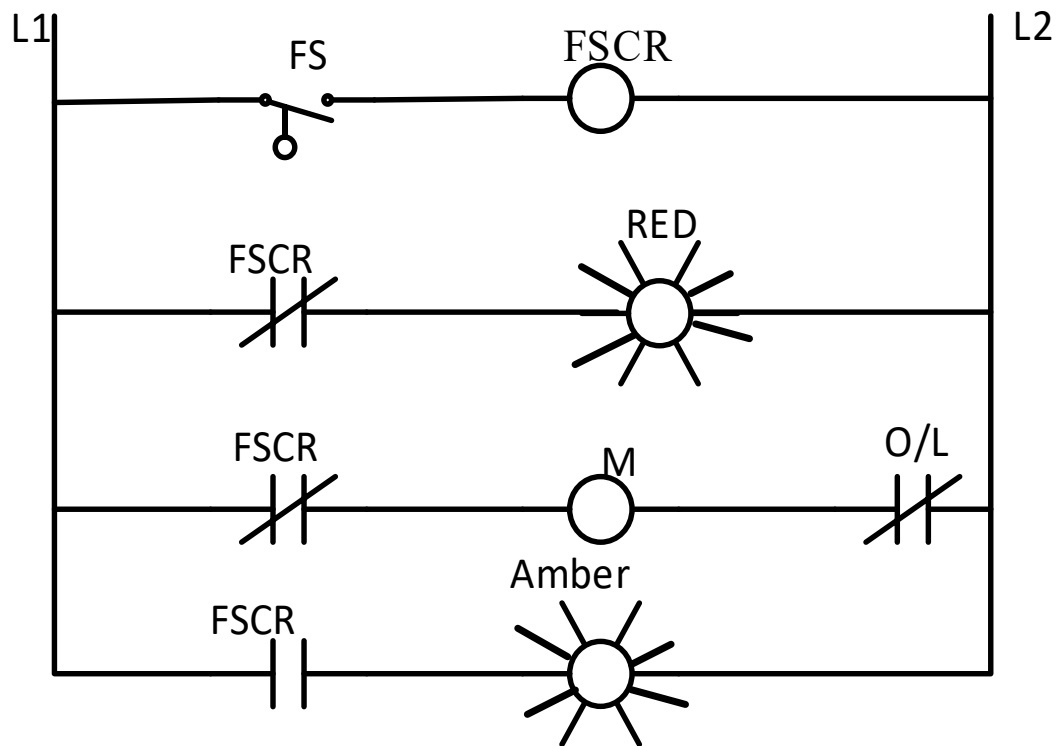


Figure 4

15. The amber lamp comes on when

- a) the overload trips **b) FS closes** c) the motor is running d) FSCR de-energizes

16. FS closes when

- a) the tank is full** b) the tank is empty c) the o/l trips d) FSCR energizes

17. The symbol labelled motor is, in reality, a/an

- a) Control relay **b) Contactor coil** c) Overload coil d) Timer relay

18. When excessive current passes through the motor

- a) the red lamp blinks b) the amber lamp blinks **c) O/L trips** d) FSCR de-energizes

19. If the control circuit at figure 11 were to be numbered, the terminals of O/L will be numbered

a)7,8

b)7,3

c)8,9

d) 6,3

20. The symbol labelled O/L is an overload

a)coil position

b)relay contact

c)main contact

d) push button

